Atomic Theory 1-2: Isotopes **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) What is an isotope?

1

Questions 2 – 5 refer to the $^{63}_{28}\text{Ni}$ isotope.

2) How many protons are contained within the nucleus?

2

3) How many neutrons are contained within the nucleus?

3

4) How many electrons are contained within a single neutral isotope of this element?

4

5) What is the mass of this isotope in amu?

5

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Questions 6 – 9 refer to the $^{238}_{92}U$ isotope.

6) How many protons are contained within the nucleus?

6

7) How many neutrons are contained within the nucleus?

7

8) How many electrons are contained within a single neutral isotope of this element?

8

9) What is the mass of this isotope in amu?

9

10) Which isotope is more likely to bond with oxygen, ^{12}C or ^{14}C? Explain.

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